HERO HOUSING FINTECH CHALLENGE · PROBLEM STATEMENT 2 · I-HACK · IIT BOMBAY E-SUMMIT 2026

गृह Griha

Automated mortgage underwriting you can explain, replay and defend to a regulator.

Home Loan and Loan Against Property decisions in about ten milliseconds: approve, reject or refer, with reasons, a Key Facts Statement in the RBI's own format, and a compliance trail built from the primary texts.

Live demo d3l1fg79dbflis.cloudfront.net
Code github.com/anshul23102/griha

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All applicant data is synthetic

THE PROBLEM

A home loan should not wait days for an answer nobody can explain.

Slow

A Home Loan or LAP file passes through people for data checks, policy checks, fraud and KYC review, property review and sign-off.

Opaque

When a decision is made, the reasons live in notes and memory. An auditor cannot replay it, and an applicant cannot learn what to change.

Heavily regulated

RBI's Fair Practices Code, Digital Lending, KYC and Responsible Business Conduct Directions and the DPDP Rules all bind every single file.

What PS2 asks for

A fully automated, regulator-compliant underwriting engine for Home Loans and LAP, covering eight capabilities: data integration, eligibility and policy, fraud and KYC/AML, risk scoring, documents and property, compliance, an instant approve / reject / refer decision, and explainability with governance.

OUR ANSWER

Griha decides the clear cases alone and shows its work on every case.

Language models read documents. They never decide. Every outcome comes from a hashed rulebook and a calibrated, monotone risk model.

10 ms
median engine time per decision on a laptop

62.5%
of 1,000 synthetic cases decided with no person involved

19
policy rules as code, plus 15 compliance controls on every case

4
regulatory texts read in their primary form, paragraph by paragraph

Explainable

Reasons, evidence behind each, a replayable record, and a Path to Yes for fixable cases.

Regulator-aware

Controls cite their paragraph and say how far we verified them. The Key Facts Statement matches RBI Annex I.

Honest

Synthetic data is labelled. We show a real-data check, the limits, and what is not built.

END TO END

One application becomes one decision through eight stations.

1 Data
Bank, bureau, KYC, land records, fetched in parallel

2 Income
Twelve months of credits become a verified income

3 Property
Valuation range, comparables, title, valuation reports

4 Eligibility
FOIR at a stress rate, LTV caps, tenure, counter-offer

5 Policy
19 hashed rules, each with a reason

6 Risk
Calibrated default probability, drivers, price

7 Fraud + KYC/AML
Tamper checks, four-list sanctions screen, linked applicants

8 Compliance
15 controls with clause and status

IN
Four-step form, a PDF bank statement, or Account Aggregator data in the ReBIT format. Consent is recorded first.

→

DECISION IN ABOUT 10 MS
An outcome, an offer, reasons, a Key Facts Statement and an audit entry holding the encrypted input.

→

OUT
APPROVE with offer · **REJECT** with reasons · **REFER** to a Credit, Fraud, Property or Data queue, evidence ready.

REQUIREMENT 1 OF 8 · INTEGRATE DATA SOURCES

Four sources feed the engine, and a missing one is never guessed.

SOURCE	CHANNEL	STATUS
Bank statement	PDF parser, or ReBIT DEPOSIT FI file (XML or JSON)	REAL PARSER
Credit bureau	Simulated, shaped like a bureau report	SIMULATED
KYC and sanctions	Simulated KYC; real UN, OFAC, EU and UK list screen; V-CIP checklist	LISTS REAL
Land records	Simulated title, RERA, encumbrance	SIMULATED

- All four are fetched **in parallel** by a Temporal workflow with retries. Every case shows each source's channel, time, record count and the stages it feeds.
- If a source stays down the case goes to the **Data queue**. Nothing is guessed. Try the *flaky bureau* test in the demo: the workflow retries and succeeds on attempt 3.
- The Account Aggregator parser reads the public ReBIT schema, rejects DTDs and entities, caps size and sanitises narrations.
- Documents are checked for balance continuity, holder match and text addressed to an AI reviewer.

REQUIREMENT 2 OF 8 · ELIGIBILITY AND POLICY

Credit policy is code: hashed, traced, and testable before it ships.

Eligibility sizing

HOUSING LOAN UP TO	MAX LTV
₹30 lakh	90%
₹30 to ₹75 lakh	80%
Above ₹75 lakh	75%
LAP (internal limit)	60%

Housing caps are RBI HFC Directions 2025, para 58. FOIR is tested at a stress rate; tenure caps 30 years (HL) and 15 (LAP). A smaller counter-offer replaces a flat no.

Policy Impact Simulator: replay 1,000 stored applications before changing a rule

FOIR CAP 55% → 65%	NOW	PROPOSED
Auto-approved	53.8%	58.8%
Referred to officers	37.5%	32.9%
Expected loss on approved	3.12%	3.42%
Realised default (synthetic truth)	3.35%	4.08%

The 19 rules live in one YAML file whose hash is stored with every decision. The simulator shows the gain and the cost together, and a fairness check per attribute with a 95% interval.

REQUIREMENT 3 OF 8 · FRAUD DETECTION AND KYC/AML

Fraud and AML checks run on every case, and the law decides what the applicant sees.

Detect

- Balance-arithmetic and edit checks on statements
- Round-trip money and window dressing
- Cash intensity and an STR signal
- Linked-applicant graph: shared phone, property, rings
- Behaviour anomaly model

Screen

31,347

entries on the real UN, US OFAC, EU and UK sanctions lists (snapshot 10 Nov 2026), names and aliases. A contradicting birth year clears a name-only match

A close match is a possible match. It goes to an officer: it is never an automatic decline (KYC Directions para 18).

Withhold

Fraud findings, sanctions matches, the STR signal and the risk category are removed from the applicant's view (KYC paras 19, 20(4), 52). Officers see everything.

WHAT THE APPLICANT IS TOLD

"Your application needs one more check by an officer. We will contact you on your registered number."

On 120 injected fraud cases (edited statements, window dressing, identity mismatch, rings) 119 were held back. The typologies are ones we wrote; real fraud differs. Aadhaar is treated as voluntary; a missing e-KYC refers to an officer.

REQUIREMENT 4 OF 8 · RISK SCORING

A monotone, calibrated model, checked on real borrowers outside our own generator.

12 inputs
FOIR, LTV, bureau score, income variability, buffer, delinquency, enquiries, bounces, age, tenure and more

→

5 bagged LightGBM
monotone constraints: worse never lowers risk

→

Isotonic calibration
so 3% means about 3%

→

Conformal range
80% band; uncertain cases refer

→

SHAP reasons + scorecard
650 points at 30:1; band A to E sets the price

Synthetic test set (720 held out)

MEASURE	VALUE
AUC · KS · Brier	0.747 · 0.414 · 0.058
Plain logistic regression AUC	0.760
Mean predicted vs actual default	7.3% vs 6.0%

The boosted model does not beat logistic regression on this synthetic data. We say so, and keep it for the monotone guarantee and SHAP.

Real borrowers (UCI, CC BY 4.0), 5-fold out of sample

DATASET	LOGIT	FREE	MONOTONE
German Credit, 1,000	0.782	0.793	**0.788**
Taiwan cards, 30,000	0.723	0.783	**0.782**

AUC. The constraints cost almost no ranking power; calibration gaps are 1.1% and 0.4%. Not Indian mortgages: it shows the recipe, not Indian accuracy.

REQUIREMENT 5 OF 8 · DOCUMENTS AND PROPERTY

Documents are verified and the property is assessed inside the same flow.

Documents

- Bank statement as a PDF, or Account Aggregator data. Both become the same object, so every downstream check works unchanged.
- Text-layer parser by default; an optional local model reads through a typed, validated schema and is cross-checked against the PDF.
- **Injection-safe:** a sentence addressed to an AI reviewer inside a document is treated as evidence of tampering and never obeyed.
- The verification table shows PAN, Aadhaar e-KYC, statement and land record with method and status.

Property

VALUATION MODEL (SYNTHETIC)	RESULT
Median absolute error	6.3%
Within 10% of true value	80.3%
80% interval actually covers	73.9%

The interval is narrower than its label; we show that, not hide it. Price levels are synthetic.

- Quantile valuation, comparables, circle rate, title, RERA, encumbrance flags.
- Loans from ₹50 lakh need two valuation reports; from ₹75 lakh two independent ones, and the lower value is used (HFC Directions Annex V).

REQUIREMENT 6 OF 8 · COMPLIANCE CHECKS

Five regulatory documents, read in their primary text, drive the controls.

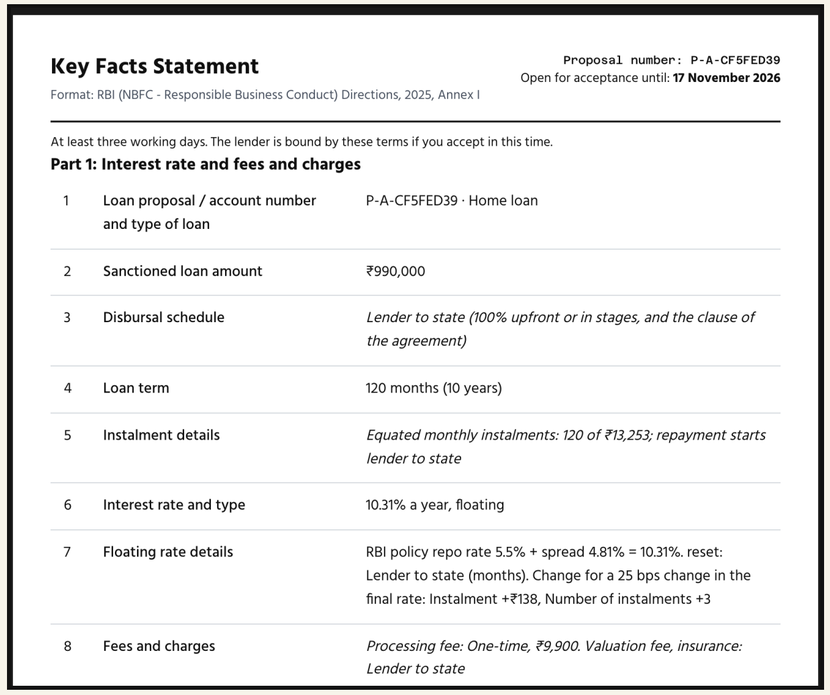
DOCUMENT	WHAT IT CHANGED IN THE PRODUCT
RBI HFC Directions 2025 Fair Practices Code, Annexes IV, V	LTV caps (para 58), KFS disclosures (131), two valuation reports, grievance route with the NHB cell (144), foreclosure (163), terms sheet
RBI Digital Lending Directions 2025	Cooling-off period, grievance escalation to RBI after 30 days, data kept in India
RBI NBFC KYC Directions 2025	Aadhaar voluntary, officer records any rejection, high-risk rating for online onboarding, tipping-off secrecy, five-year retention
RBI Responsible Business Conduct 2025	Annex I Key Facts Statement, 3-working-day validity, penal charge is not interest (para 30), prepayment (34)
DPDP Rules 2025 G.S.R. 846(E)	Privacy notice, safeguards, data-principal rights with a 90-day clock, breach register with the 72-hour deadline

"NHB regulations": since August 2019 the RBI regulates housing finance companies; its 2025 HFC Directions are the operative text. The NHB complaint cell still appears in the borrower's letter.

Reading found real defects, now fixed: the applicant view leaked sanctions findings; PAN or sanctions failures auto-declined; erasure ignored the five-year retention duty.

KEY FACTS STATEMENT

The offer comes with a statement in the RBI's own Annex I layout, in English and Hindi.



Key Facts Statement

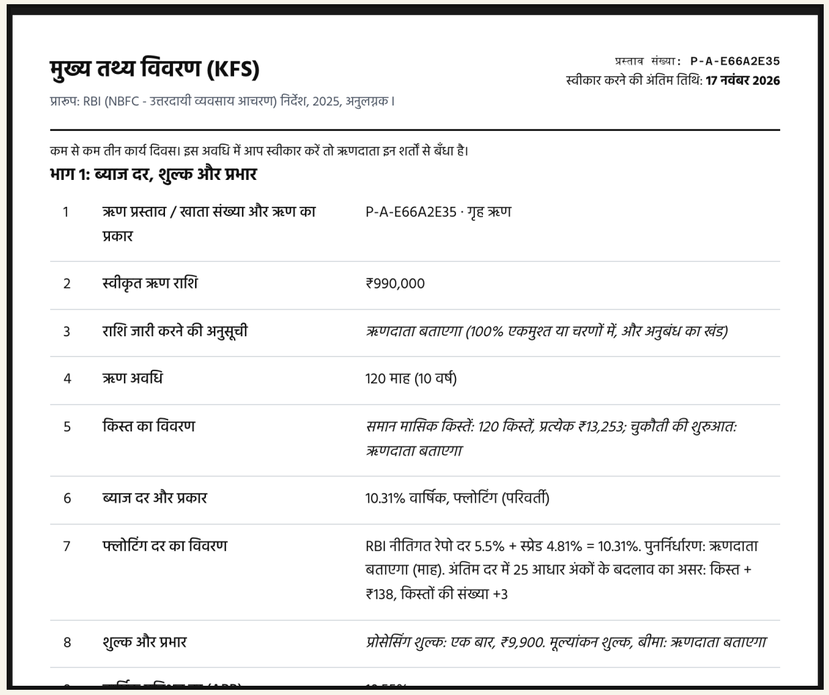
Proposal number: P-A-CF5FED39
Open for acceptance until: **17 November 2026**

Format: RBI (NBFC - Responsible Business Conduct) Directions, 2025, Annex I

At least three working days. The lender is bound by these terms if you accept in this time.

Part 1: Interest rate and fees and charges

1	Loan proposal / account number and type of loan	P-A-CF5FED39 · Home loan
2	Sanctioned loan amount	₹990,000
3	Disbursal schedule	*Lender to state (100% upfront or in stages, and the clause of the agreement)*
4	Loan term	120 months (10 years)
5	Instalment details	*Equated monthly instalments: 120 of ₹13,253; repayment starts lender to state*
6	Interest rate and type	10.31% a year, floating
7	Floating rate details	RBI policy repo rate 5.5% + spread 4.81% = 10.31%. reset: Lender to state (months). Change for a 25 bps change in the final rate: Instalment +₹138, Number of instalments +3
8	Fees and charges	*Processing fee: One-time, ₹9,900. Valuation fee, insurance: Lender to state*



मुख्य तथ्य विवरण (KFS)

प्रस्ताव संख्या: P-A-E66A2E35
स्वीकार करने की अंतिम तिथि: **17 नवंबर 2026**

प्रारूप: RBI (NBFC - उत्तरदायी व्यवसाय आचरण) निर्देश, 2025, अनुलग्नक I

कम से कम तीन कार्य दिवस। इस अवधि में आप स्वीकार करें तो ऋणदाता इन शर्तों से बँधा है।

भाग 1: ब्याज दर, शुल्क और प्रभार

1	ऋण प्रस्ताव / खाता संख्या और ऋण का प्रकार	P-A-E66A2E35 · गृह ऋण
2	स्वीकृत ऋण राशि	₹990,000
3	राशि जारी करने की अनुसूची	*ऋणदाता बताएगा (100% एकमुश्त या चरणों में, और अनुबंध का खंड)*
4	ऋण अवधि	120 माह (10 वर्ष)
5	किस्त का विवरण	*समान मासिक किस्तें: 120 किस्तें, प्रत्येक ₹13,253; चुकौती की शुरुआत: ऋणदाता बताएगा*
6	ब्याज दर और प्रकार	10.31% वार्षिक, फ्लोटिंग (परिवर्ती)
7	फ्लोटिंग दर का विवरण	RBI नीतिगत रेपो दर 5.5% + स्प्रेड 4.81% = 10.31%. पुनर्निर्धारण: ऋणदाता बताएगा (माह). अंतिम दर में 25 आधार अंकों के बदलाव का असर: किस्त + ₹138, किस्तों की संख्या +3
8	शुल्क और प्रभार	*प्रोसेसिंग शुल्क: एक बार, ₹9,900. मूल्यांकन शुल्क, बीमा: ऋणदाता बताएगा*

- Proposal number, and validity of at least three working days
- Part 1 rate and fees, Part 2 qualitative items
- APR computation sheet and the full repayment schedule
- Effect of a 25 bps rate change
- Borrower acknowledgement recorded in the audit log

17.07%

Our APR reproduces the RBI's worked example: ₹20,000 at 15% for 24 months with ₹400 fees. It is a permanent test.

Items only a lender can know (penal-charge amount, reset frequency, grievance officer) read "Lender to state". We never invent them. The Hindi text awaits a legal translator's review.

REQUIREMENT 7 OF 8 · INSTANT DECISIONING

Clear cases are decided alone. A person sees only what the law or the risk requires.



1,000 synthetic applicants through the live engine, median 10 ms of compute. Green is approve, red is reject (8.7%), amber is refer.

Where the humans come in

QUEUE	SHARE OF CASES
Credit (grey zone, over-ask, bureau band)	25.5%
Fraud and compliance	5.1%
Property (title, plan, valuation)	4.9%
Data (a source was down)	2.0%

We tried loosening the limits

ELIGIBILITY THRESHOLD	AUTOMATIC	DEFAULT OF APPROVED
0.60 (shipped)	62.5%	3.4%
0.50	66.4%	4.2%
0.40	69.8%	4.9%

The automatic rate rises, and so do defaults in the simulation. We left the limits; a lender sets them from its own losses.

REQUIREMENT 8 OF 8 · EXPLAINABILITY AND GOVERNANCE

Every decision is explainable, traceable and audit-ready.

1

Explain

- Reasons on every outcome, each with its evidence
- SHAP drivers and a scorecard number
- **Path to Yes:** the smallest change that flips a no
- Printable decision letter, English or Hindi

2

Trace

- Hash-chained audit log; the input is stored encrypted
- Replay any decision byte for byte
- Tamper detection, shown live
- Engine and policy version on every decision; a changed engine is flagged, not mistaken for tampering

3

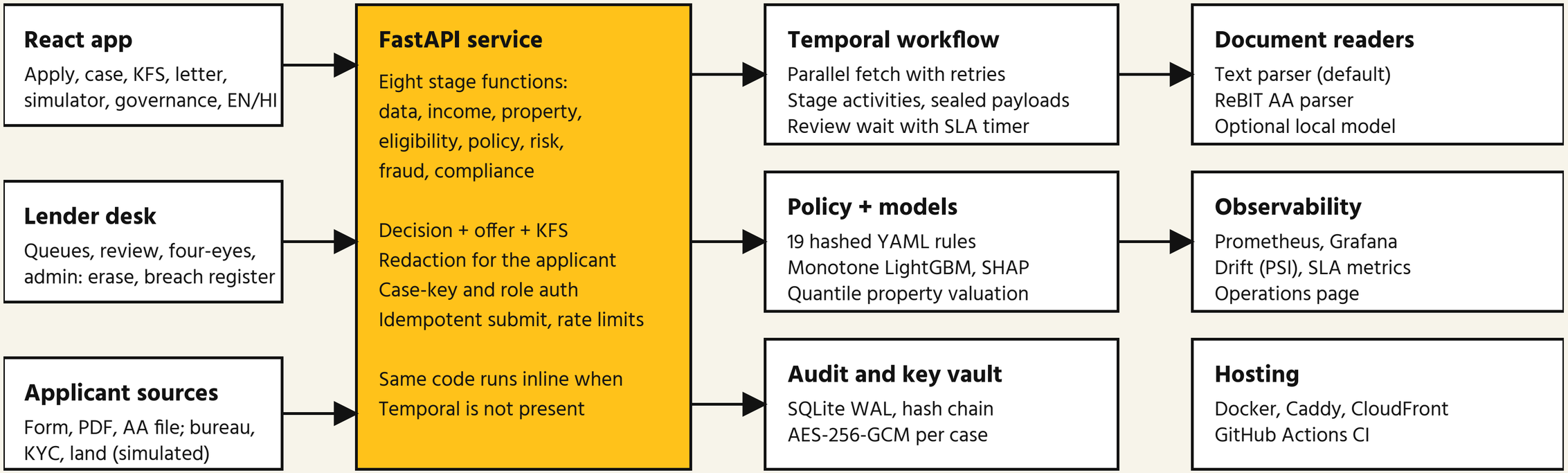
Govern

- Four-eyes: an officer cannot approve a fraud case; the Credit Head can
- Referrals carry an SLA timer that escalates and audits
- Model card, drift monitor, fairness check per attribute
- Erasure by crypto-shredding; the chain still verifies

Replay, tamper detection, retention and redaction are all covered by automated tests (149 backend, 29 browser including accessibility checks).

ARCHITECTURE

One stateless engine, a durable workflow around it, and an audit log as source of truth.



SECURITY AND PRIVACY

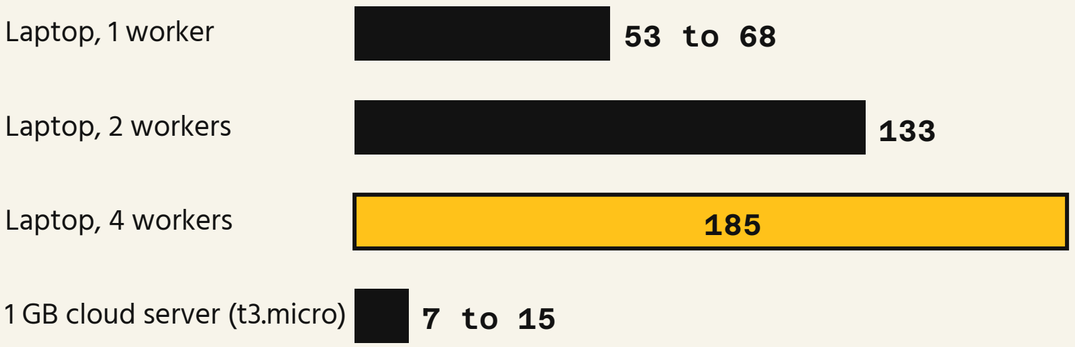
Built as if the data were real, because controls should not depend on fake data.

ASSET	CONTROL
Applicant data	AES-256-GCM envelope encryption, one data key per case; the audit log stores ciphertext; the workflow engine holds sealed payloads too
Who can open a case	A per-case capability key in the link fragment, or a staff login. A link without its key is a plain "not found"
What an applicant sees	Fraud, sanctions, STR and risk category removed (tipping-off rules); officers' notes stay internal
Rights and retention	Withdraw consent, access, correct, erase with a 90-day clock; a retention hold defers erasure where KYC requires records; breach register with the 72-hour deadline
Inputs	Strict schemas, errors never echo input, control and bidi characters refused, XML entities rejected, size caps, prompt-injection treated as evidence
Access	Salted PBKDF2 (600k), lockout per visitor and account, rate limits on the real client address behind the CDN, a private admin password on the public demo
Supply chain	bandit, pip-audit, pnpm audit and a container build in CI on every push; all clean

SCALABILITY AND OPERATIONS

Measured throughput, with the limits stated.

Decisions per second (synthetic cases)



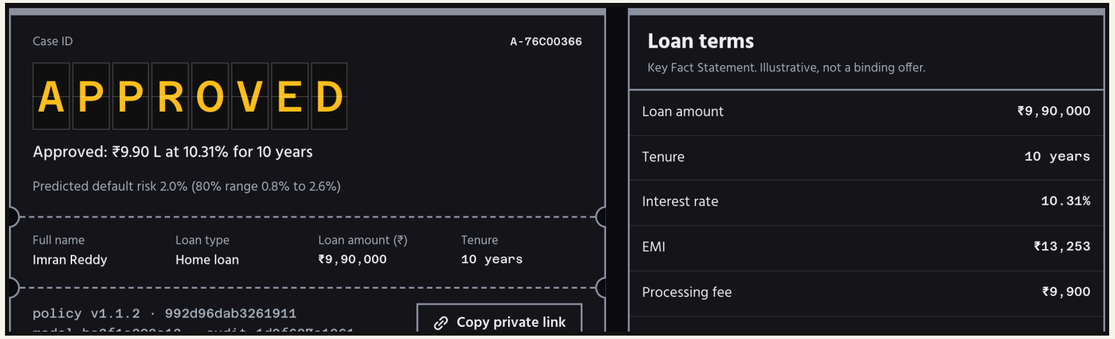
Zero errors, one intact hash chain, one key per case.

- A 120-second sustained run on the laptop completed **13,718 decisions with no errors**, using 20 KB of storage per decision.
- The load test found three real bugs (lost keys, startup races, private memory). All fixed and tested.
- Stateless stage functions; the audit log is the shared state, so more worker processes scale it almost linearly on one machine.
- Temporal gives retries, durable review timers and a history; Prometheus and Grafana show decisions, stage latency, retries, drift.

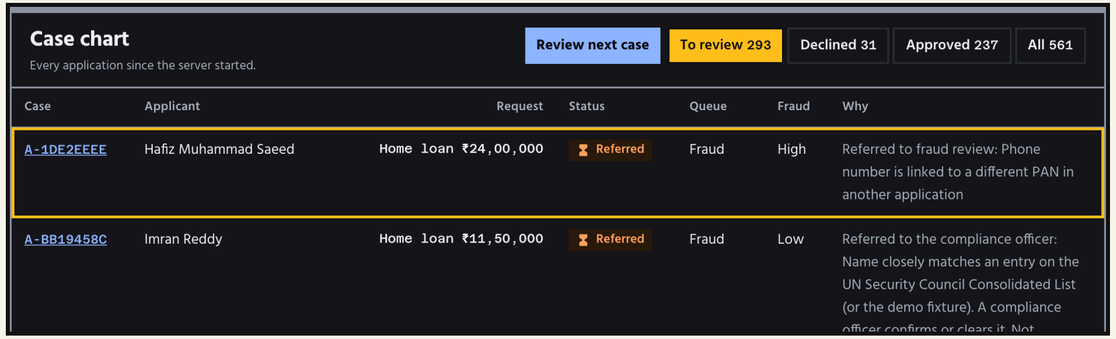
Not claimed: a million decisions a day. That is 11.6 a second on average; peaks, bureau latency and the database would decide it. docs/scalability.md shows the arithmetic.

THE PRODUCT

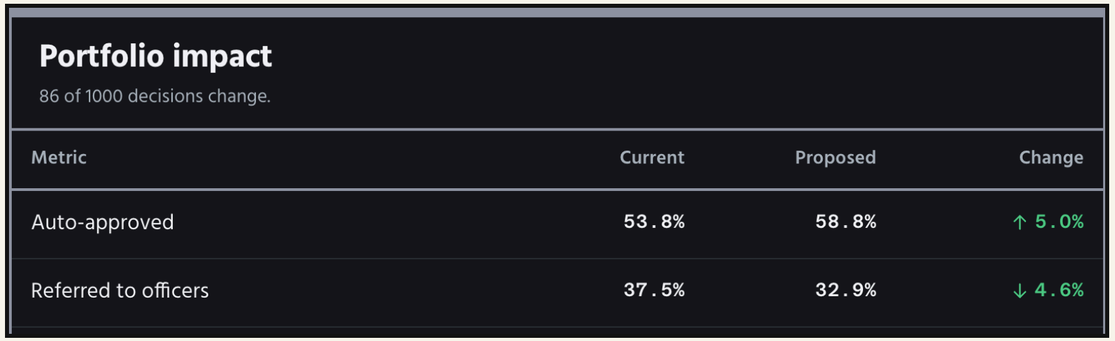
A bold, bilingual interface, in English and Hindi, night and day, designed like Indian Railways signage.



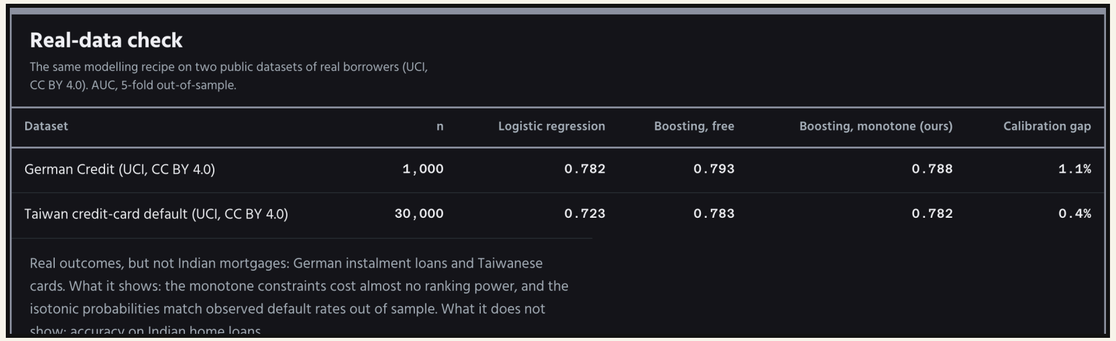
Case: verdict ticket, route strip and loan terms, with a private link.



Lender desk: referred cases with queue, fraud level and the reason.



Policy simulator: change a threshold, see approvals, loss and default together.



Governance: the real-data check beside audit, replay and the model card.

HONESTY

What is real, what is simulated, and what is not built.

REAL

- APR (matches the RBI example); KFS in English and Hindi
- UN, OFAC, EU and UK list screening (31,347 entries)
- Quarterly statement, document release, recovery rules
- AA signing and encryption (JWS, AES-GCM)
- V-CIP rules (paras 26, 27); PostgreSQL support
- Repo rate 5.50% and RESIDEX figures
- Hash-chained audit and encryption
- DPDP rights flow, breach register

SIMULATED

- Bureau, KYC and land-record feeds
- Every applicant and bank statement
- The Aggregator itself: a mock, not a registered one
- V-CIP sessions: no real video, camera hashes only
- Property price levels
- Default outcomes the model is tested on

NOT BUILT

- Registered FIU and live Aggregator (needs RBI and Sahamati onboarding)
- MHA UAPA list (no machine-readable feed); PEP data (licensed)
- CKYCR upload (needs CERSAI access)
- V-CIP video recording and liveness model
- DSA/DMA rules; KFS summary box
- India-region hosting (AWS free plan locks to Sydney)

LIMITS AND NEXT STEPS

What a lender must do before this makes a real decision.

Known limits

- **No Indian loan-level data** we could find. Our synthetic default rate is several times the real housing NPA ratio (about 0.5% for banks at March 2026, from press reports of the RBI's June 2026 report, not checked in the primary text).
- Policy cut-offs (FOIR, age, bureau, default) are illustrative; only the LTV caps come from the Directions.
- The Hindi text needs a legal translator; the demo runs in Sydney on synthetic data.

What we built to close the gap

- **Bring your own data:** one command, `griha.byod --csv book.csv`, validates a lender's loan book, retrains the same monotone calibrated model, reports out-of-sample AUC, KS and calibration, and writes a separate model. The shipped model is never overwritten.
- **Hindi review kit** and a sign-off file; the app only drops the "unreviewed" notice when a named reviewer signs.
- Steps to move hosting to India, and the cut-offs to be re-set with the Policy Impact Simulator.

SOURCES AND CITATIONS

Every figure the product relies on has a source, a date and a status.

SOURCE	USED FOR	STATUS
RBI Housing Finance Companies Directions, 2025 (Ch. X; Annexes IV, V)	LTV caps, KFS disclosures, valuation reports, grievance route	PRIMARY TEXT
RBI Digital Lending Directions, 2025	KFS, cooling-off, grievance, data localisation	PRIMARY TEXT
RBI NBFC KYC Directions, 2025 (RBI/DOR/2025-26/361)	KYC, AML, tipping-off, retention	PRIMARY TEXT
RBI NBFC Responsible Business Conduct Directions, 2025 (.../362)	Annex I KFS, penal charges, prepayment, APR example	PRIMARY TEXT
Digital Personal Data Protection Rules, 2025 (G.S.R. 846(E), 892(E), 843(E))	Notice, safeguards, rights, breach, retention	PRIMARY TEXT
RBI policy repo rate 5.50% (MPC, Oct 2026); NHB RESIDEX Q3 FY2025-26	Pricing base rate; market context	PUBLISHED
UN, OFAC SDN, EU and UK sanctions lists (10 Nov 2026)	Sanctions screening	PUBLIC LIST
ReBIT Account Aggregator DEPOSIT FI schema	Statement file format	PUBLIC SPEC
UCI German Credit (doi:10.24432/C5NC77); Taiwan card default (doi:10.24432/C55S3H), CC BY 4.0	Real-data check of the model recipe	OPEN DATA
RBI Financial Stability Report, June 2026 (press reports)	Sanity check on the real housing NPA ratio	SECONDARY

Freddie Mac loan-level data was used only for a private local check; it is not part of the product and no result is published. Links: the Data sources page in the app.

OPEN IT AND JUDGE FOR YOURSELF

Run a case. Open the reasons. Replay the decision.

LIVE DEMO
d3l1fg79dbflis.cloudfront.net

SOURCE CODE, TESTS, DOCS
github.com/anshul23102/griha

A two-minute tour

1. Run a live case on the home page and open the full case.
2. Open the Key Facts Statement and switch to Hindi.
3. Apply with the flaky-bureau test and watch the retry.
4. Sign in as officer@griha.demo (password griha-demo) and review a referral.
5. In Governance, verify the audit chain and replay a decision.

All applicant data is synthetic. Model metrics describe a simulation. This is a prototype, not a lending product or legal advice.